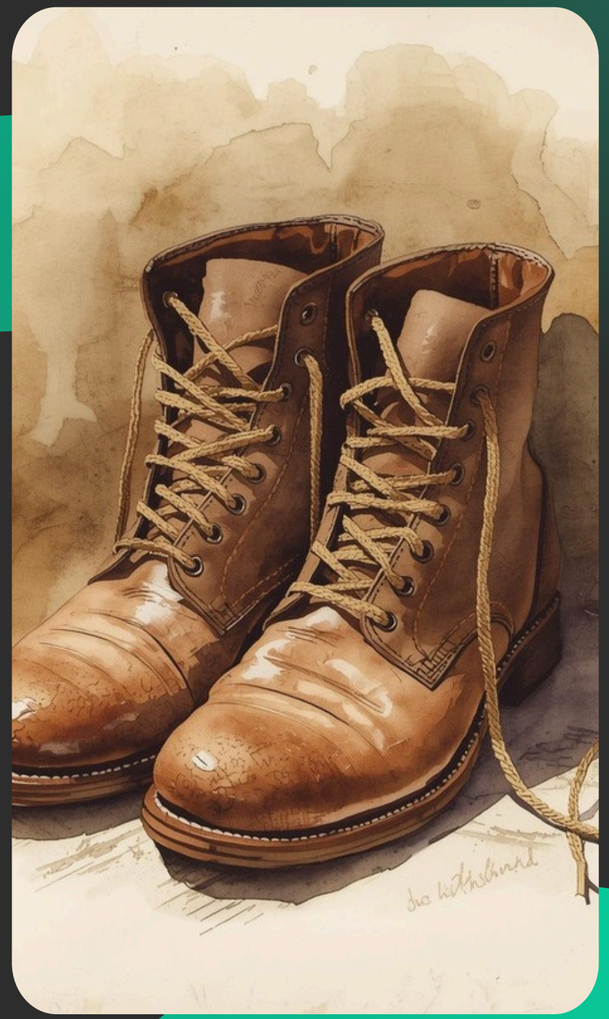
Step Forward

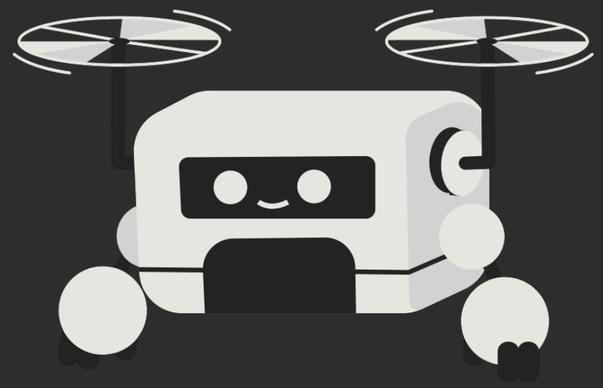
AI-Powered Footwear App

Presented By:
Aryan Chakraborty



Problem Statement

- **Manual distributor search is slow, inconsistent, and lacks insights.**
- Create AI agent that finds, analyzes, and ranks distributors using public data.



Solution Approach

An end-to-end AI sales agent built with LangChain, LangGraph, FastAPI and Streamlit to help a footwear brand identify, qualify, and engage global distributors.

Target Audience

Who Benefits

The application is catered towards

- Footwear brands looking for distributors
- Distributors looking for well performing brands
- ERP systems for companies
- Investors interested in companies

How the Agent Works

1. The agent is built as a stateful multi-node pipeline using LangGraph, where each node performs a distinct sales task
2. The pipeline runs end-to-end from lead discovery to CRM-style action recommendations, with a separate conversational interface layered on top using Langchain and LLM's
3. The backend is served via FastAPI, which exposes the pipeline as REST endpoints.
4. The frontend is a Streamlit web app with six tabs — Upload, Prospect, Qualify, Outreach, Next Action, and Chat — each mapping directly to a stage of the sales workflow.

Tech Stack and Flowchart

- LLM:

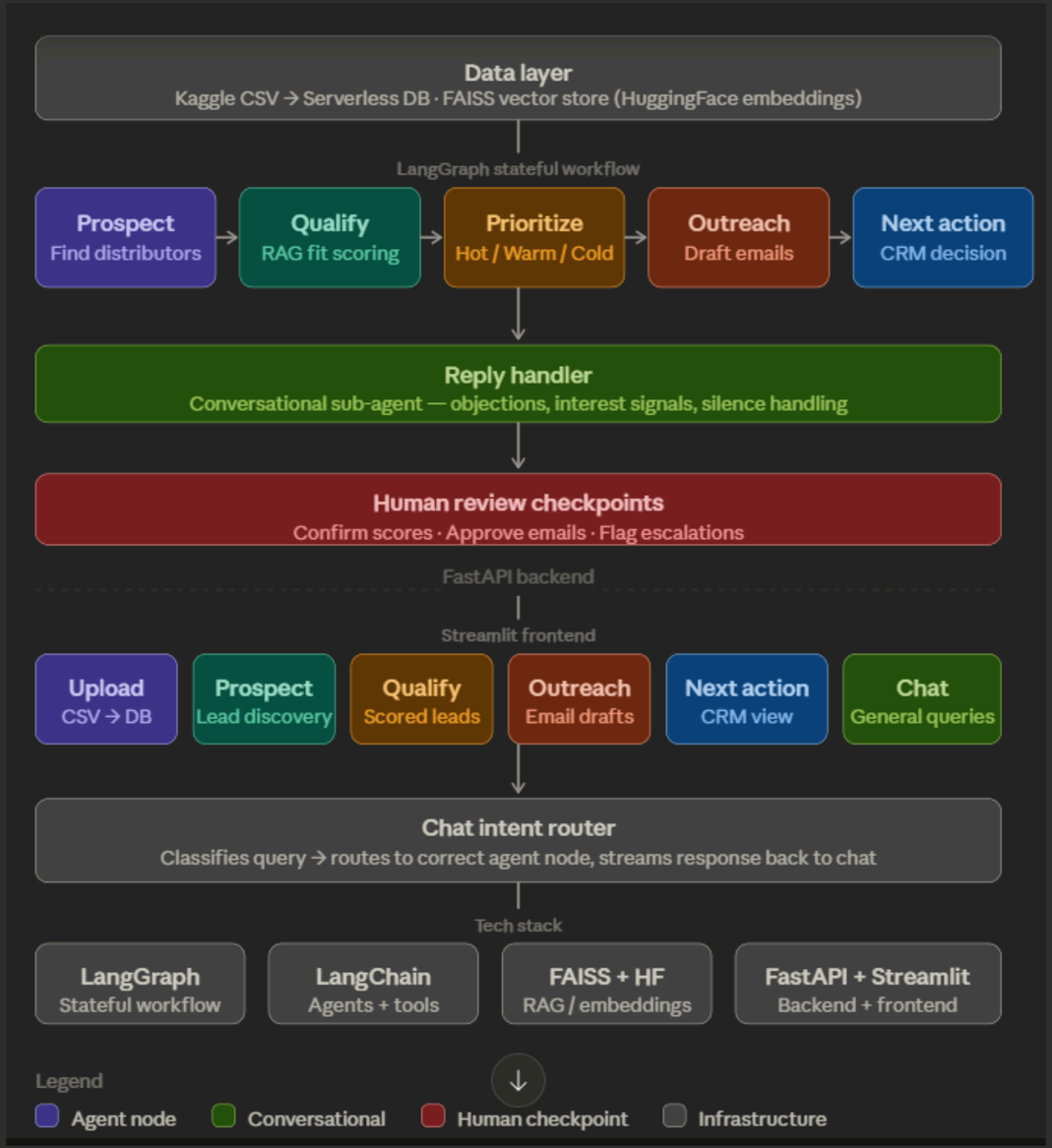
Zephyr-7B-Beta via HuggingFace Router (featherless-ai provider)

- Embeddings:

MiniLM-L6-v2 (local ~90MB)

- Stack:

LangGraph
LangChain 0.2+
FastAPI
RAG Vectorstore
Streamlit + Ngrok (Application)



How Leads Are Found and Qualified

Lead discovery (prospect_node) loads dealer profiles from the database.

In this demo the database is populated from the Kaggle footwear wholesale dataset — 3054 transaction rows aggregated into 7 dealer profiles capturing revenue, margin, product categories, brands carried, stock levels, and location.

Qualification (qualify_node) applies a two-layer scoring approach:

- Rule-based scoring evaluates five dimensions — years active, category fit, margin compatibility, revenue range, and stock management capability — each weighted. Category fit carries the highest weight (30%)
- Pool-relative normalisation scales each dealer's score against the min and max within the current pool
- LLM narrative — RAG context is retrieved and passed to Zephyr to generate a 2-3 sentence qualitative assessment per dealer, surfacing nuance that the rule scores cannot capture.

How the Agent Decides Who to Prioritise

Prioritisation (prioritize_node) uses percentile-based tier assignment rather than fixed thresholds. The top 25% of scored dealers are flagged Hot, the middle 50% Warm, and the bottom 25% Cold. This approach guarantees a realistic distribution regardless of how similar or different the dealer pool is.

Condition	Recommended Action
Hot tier, no reply yet	Schedule Meeting
Hot tier, score ≥ 88/100	Escalate to Senior BDM
Warm tier	Follow-up in 5 days
Cold tier	Deprioritise — review in 6 months
Any tier, positive reply received	Override to Schedule Meeting
Any tier, negative reply received	Override to Deprioritise

Where Human Review Is Needed

Three explicit checkpoints are built into the workflow where the agent pauses for human judgement:

1. After Qualification

2. Before Outreach is Sent — every generated email is presented as an editable draft.

3. Escalation Flagging — dealers scoring above 88/100 are automatically flagged for senior BDM involvement rather than handled by a junior rep or the agent alone. This ensures high-value opportunities get appropriate attention and are not lost to under-resourced follow-up.

Thank You Very Much

aryanchaks3602@gmail.com